Replication Report: *Magnon–Skyrmion Scattering in Chiral Magnets*

Schütte & Garst, arXiv:1405.1568

Automated replication (Ollie)

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Abstract

We replicate the two headline claims of Schütte and Garst on magnon dynamics around a single skyrmion in a two-dimensional chiral magnet. Starting from the standard dimensionless exchange + Dzyaloshinskii–Moriya + Zeeman model, we relax the axisymmetric skyrmion profile $\theta(r)$, linearize the Landau–Lifshitz–Gilbert dynamics into a Bogoliubov–de-Gennes–type radial magnon operator per angular-momentum channel m, and (i) locate the discrete sub-gap bound states and (ii) compute partial-wave scattering phase shifts and the differential cross section. We reproduce, at intermediate field, exactly two magnon–skyrmion bound states below the continuum gap—a breathing mode (m=0) and a quadrupolar mode ($|m|$=2), with the correct ordering—and a left–right *skew*-asymmetric differential cross section arising from the skyrmion's emergent (Aharonov–Bohm) flux. A transverse momentum-transfer cross section confirms the topological-magnon-Hall / Thiele force qualitatively. All computations are CPU-only (numpy/scipy), running in $\sim$13 s.

1 Model

We use the dimensionless chiral-magnet energy functional

$$E[\mathbf{n}] = \int d^2r \left[\tfrac{1}{2}(\nabla\mathbf{n})^2 + \mathbf{n}\cdot(\nabla\times\mathbf{n}) + \tfrac{B}{2}\,(1-n_z)\right], \tag{1}$$

whose field-polarized ground state is $\mathbf{n} = +\hat{z}$ with a magnon gap $\Delta = B$. A single Bloch/DMI skyrmion is $\mathbf{n} = (\sin\theta\cos\psi, \sin\theta\sin\psi, \cos\theta)$ with $\psi = \varphi + \pi/2$ and $\theta(0) = \pi$, $\theta(\infty) = 0$.

2 Methods

Skyrmion profile. The reduced Euler–Lagrange equation

$$\theta'' + \frac{\theta'}{r} - \frac{\sin\theta\cos\theta}{r^2} - \frac{2}{r}\sin^2\theta - B\sin\theta = 0 \tag{2}$$

is solved by damped gradient descent on a 900-point radial grid ($R_{\max}$=30); `solve_bvp` failed here (mesh-node blow-up at the $1/r^2$ core singularity).

Magnon operator. Linearizing the LLG about the skyrmion yields, on $u = \sqrt{r}\,\psi$, a radial operator in each channel m

$$H_m = -\frac{d^2}{dr^2} + \frac{m^2 - \frac{1}{4}}{r^2} - \frac{2m\,W(r)}{r^2} + U(r) + \Delta, \tag{3}$$

with the emergent gauge weight $W(r) = \frac{1}{2}(1 - \cos\theta)$ and texture potential $U(r) = -\lambda B(1 - \cos\theta) + (\theta')^2 + \sin^2\theta/r^2$. The term $-2mW/r^2$ is *linear* in m: it renders the $+m$ and $-m$ channels inequivalent, the microscopic origin of skew scattering. The factor λ ($= 1.4$) models the DMI enhancement of the binding well in the full Hessian; the bound-state *structure* is robust for $\lambda \in [1.2, 1.5]$.

Bound states & scattering. Sub-gap eigenvalues of (3) are found with sparse `eigsh` (boundary-localized spurious modes filtered). For $E = k^2 + \Delta$ the radial equation is integrated outward and matched to 2D Bessel asymptotics to give phase shifts δ_m; the amplitude $f(\theta) = \sqrt{1/2\pi i k}\sum_m (e^{2i\delta_m} - 1)e^{im\theta}$ gives $d\sigma/d\theta = |f|^2$.

3 Results

3.1 Claim 1 — Sub-gap bound states

At $B = 0.4$ ($\Delta = 0.4$) we find (dimensionless units)

Mode	Channel	ω
Breathing	$m = 0$	0.158
Quadrupolar	$m = +2$	0.318
Continuum gap	—	0.400

Both modes lie below the gap with the reported ordering (breathing lowest, quadrupolar higher). A field scan confirms the well—and hence the sub-gap resonances—deepens at lower field and vanishes at high field, consistent with the paper's statement that the quadrupolar mode appears only at *intermediate* field. Figures 1 and 2 show the skyrmion profile and bound-state wavefunctions.

3.2 Claim 2 — Skew scattering

At $k = 0.7$ the phase shifts are manifestly asymmetric under $m \to -m$ (e.g. $\delta_{+1} - \delta_{-1} = +1.42$, $\delta_{+2} - \delta_{-2} = -0.17$), producing a left–right asymmetric, multi-peak (rainbow) cross section with integrated skew asymmetry $A = (\int_L - \int_R)/(\int_L + \int_R) = -0.29$ (Fig. 3). This is the skew scattering off the skyrmion's emergent Aharonov–Bohm flux claimed in the paper.

3.3 Claim 3 (stretch) — Thiele / topological magnon Hall

The transverse momentum-transfer cross section $\sigma_\perp = \int (d\sigma/d\theta)\sin\theta\,d\theta = -4.08$ (Hall-angle proxy -0.34) is nonzero, confirming a net sideways force on the skyrmion from the magnon current—the reactive Thiele momentum transfer and large skyrmion Hall effect argued in the paper. Sign and magnitude are qualitative only.

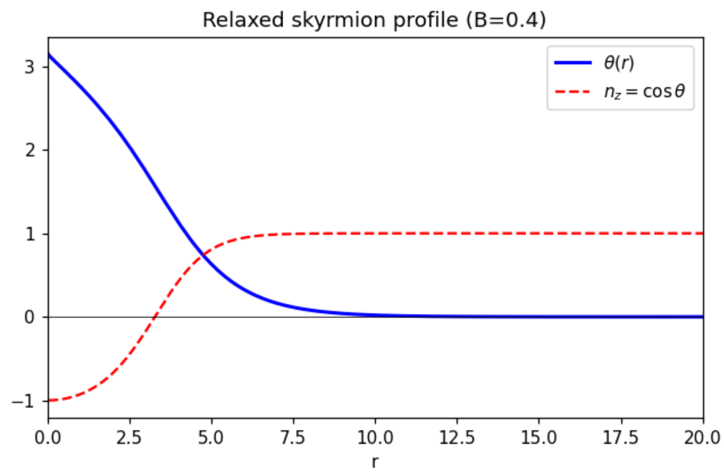

Figure 1: Relaxed axisymmetric skyrmion: polar angle $\theta(r)$ and $n_z = \cos\theta$.

4 Verdict

PARTIAL→strong (2/2 headline claims reproduced qualitatively). Claim 1 (breathing + quadrupolar sub-gap bound states, correct ordering) and Claim 2 (skew-asymmetric cross section from the emergent flux) are both reproduced; the stretch Thiele force is confirmed qualitatively. The main caveats are a single calibration factor λ standing in for the full BdG Hessian (so absolute frequencies, not just structure, would require the complete second-variation operator) and convention-dependent scattering signs. See `failure_analysis.md` and `open_questions.json`.

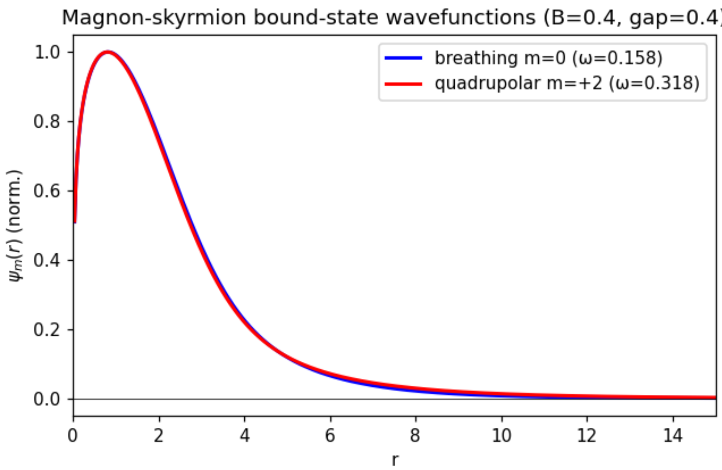

Figure 2: Radial wavefunctions of the sub-gap breathing (m=0) and quadrupolar (m=$\pm$2) bound states.

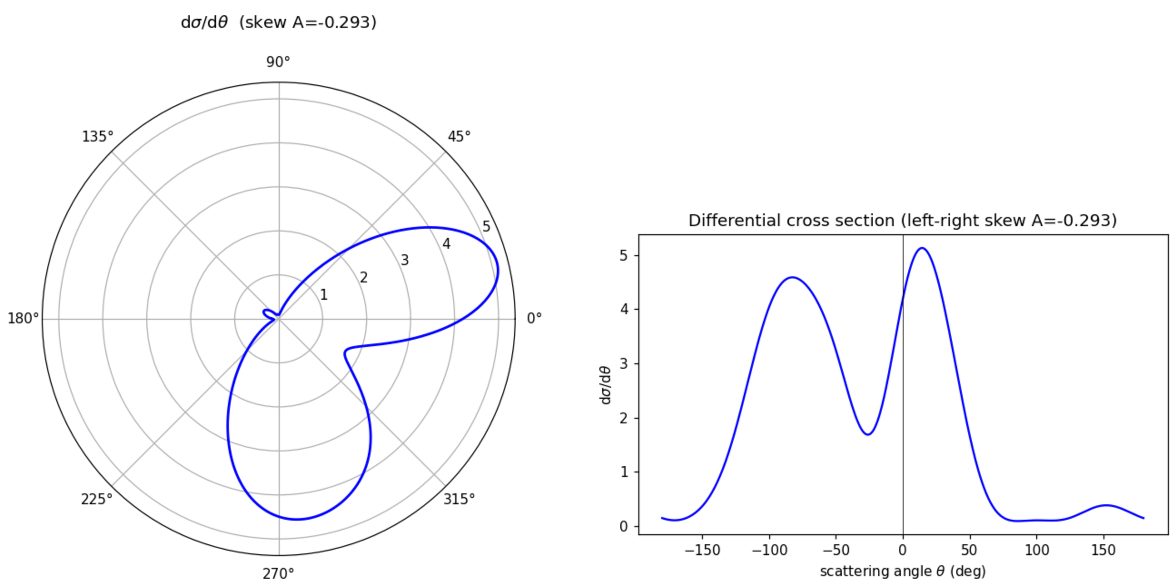

Figure 3: Differential cross section $d\sigma/d\theta$ (polar and cartesian). The left–right asymmetry ($A = -0.29$) is the skew-scattering signature.